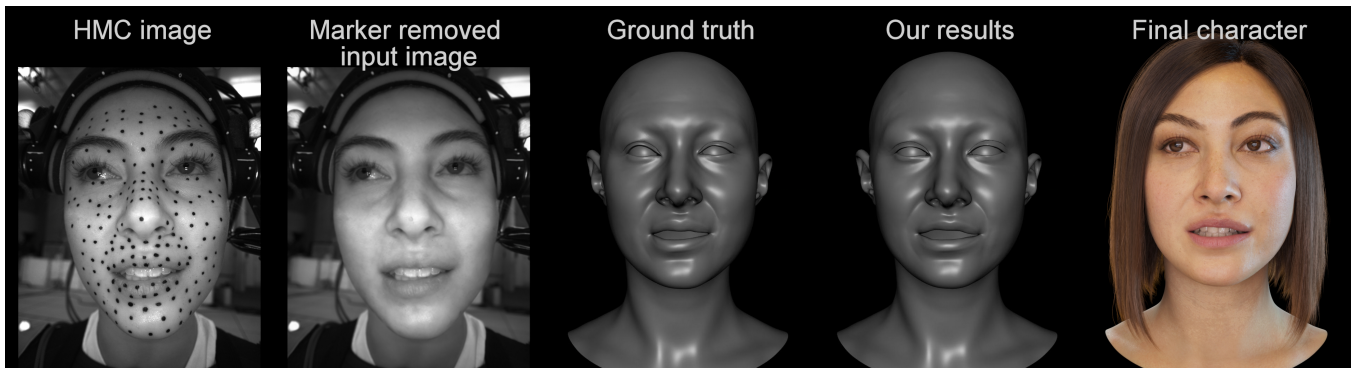


# Accelerating facial motion capture with video-driven animation transfer

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**Figure 1: Results of our approach. From left to right: Raw HMC captured image; marker-removed input image; ground truth from offline motion capture system [Moser et al. 2017]; our results; our results in a finalised character render.**

## ABSTRACT

We describe a hybrid pipeline that leverages: 1) video-driven animation transfer [Moser et al. 2021] for regressing high-quality animation under partially-controlled conditions from a single input image, and 2) a marker-based tracking approach [Moser et al. 2017] that, while more complex and slower, is capable of handling the most challenging scenarios seen in the capture set. By applying the most suited approach to each shot, we have an overall pipeline that, without loss of quality, is faster and has less user intervention. We also improve the prior work [Moser et al. 2021] with augmentations during training to make it more robust for the Head Mounted Camera (HMC) scenario. The new pipeline is currently being integrated into our offline and real-time workflows.

## CCS CONCEPTS

• Computing methodologies → Motion capture.

## KEYWORDS

Face Capture, Animation, Optimization, Head Mounted Camera

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## 1 INTRODUCTION AND OVERVIEW

Production-level facial motion capture traditionally relies on tracking-based techniques that focus on facial features. Marker-based approaches [DD 2022; Weta 2022] are particularly common and, by relying on salient features, work well even in challenging lighting situations. There are also high quality markerless solutions [Disney 2022], generally working in more controlled scenarios. These techniques can, however, suffer from registration and sliding issues, caused e.g. from marker changes due to reapplication or optical flow failing to match pore details between frames. Fixing these requires manual or semi-automatic steps which make the process slower and more cumbersome. Another group of techniques, based directly on regressing the animation from images [Klaudiny et al. 2017; Moser et al. 2021], has recently shown promising results, although their use in a production scenario has remained limited. These skip tracking of facial features, which makes them faster to apply and consistent between shots. This comes at the cost of being heavily reliant on the training data that can be limited/costly to obtain. They also tend to be less user controllable, lacking a clear mapping between features, e.g. markers or skin pores, and the mesh.

We present a hybrid pipeline that combines a more traditional marker based approach [Moser et al. 2017] with the regression, video-driven animation transfer technique of Moser et al. [2021]. For shots more closely resembling the training data and with less extreme occlusions and lighting variations, we use an improved version of the latter that has been tailored to HMC. For more challenging shots, we fall back to Masquerade [Moser et al. 2017]. We capture all shots with markers, as that is required for Masquerade

and we do not know which approach will be used in each shot. This would, however, impact negatively our regressor due to the inconsistencies caused by markers. We thus remove markers from the plates using a technique similar to Zhu et al. [2017]. We train the regressor with marker-removed HMC images and animation data obtained with e.g. Di4D [2022] or Masquerade, thus creating a model that predicts animation weights directly from an image. As Masquerade is employed less frequently, the overall pipeline is simpler and faster to use. Additionally, the shots processed with Masquerade can help to further generalise our regressor.

Our pipeline relies on both Moser et al. [2021; 2017], so we forward the reader to these for more details. The remainder of the abstract focuses on the improvements to Moser et al. [2021] required for the HMC scenario. To this end, we improved the segmentation masks, introduced background augmentations, and infer the results using a locked frame alignment. All these made the model more robust to variations seen in capture and led to less noisy results.

The main advantages of our hybrid pipeline are: 1) significant speed-up over a traditional marker-based approach, without loss quality; 2) combination of regression and tracking based techniques with no changes of the capture pipeline; 3) extension of Moser et al. [2021] by making it more robust and stable in an HMC scenario.

## 2 VIDEO-DRIVEN TRANSFER FOR HMC

To train the regressor [Moser et al. 2021], we capture a small number of training shots, i.e. range of motion and visemes, in the HMC and track them with Masquerade. From these, we generate a PCA basis to obtain their *animation coefficients*. We additionally render the sequences as if observed from the HMC cameras. The markers are removed using an auxiliary image-to-image translation model [Zhu et al. 2017], which has been pre-trained on multiple identities captured with and without facial markers. All sequences are processed as described in Moser et al. [2021], except for in the CG domain. Here we generate pixel-perfect masks using 3D geometry. This led to less noise when regressing unseen shots.

### 2.1 Training

Moser et al. [2021] showed promising results in a controlled scenario, however, HMC recordings can have noticeable differences between the training and unseen sequences. One of the most significant is the background, which inevitably changes as the person moves through the capture volume. We could overfit the model to each shot, but our goal is a generalized model to speed up processing of new shots. We thus sample random rectilinear crops from several indoor panoramic high dynamic range images (HDRI), at multiple exposures, and compose them with each sample, using its processed segmentation mask. Finally, we use views from both cameras to make the model more robust to pose variations. Training, otherwise, remains the same as Moser et al. [2021].

### 2.2 Inference

Inference happens on marker-removed plates from a single camera. We choose the camera that best frames the actor. New shots follow the same processing steps as the training data, however, we compute the frame region only once and use it consistently through all the sequences. This produces less noisy results than

when computing this region for each frame, which is particularly visible in consecutive frames.

## 3 RESULTS/VALIDATION

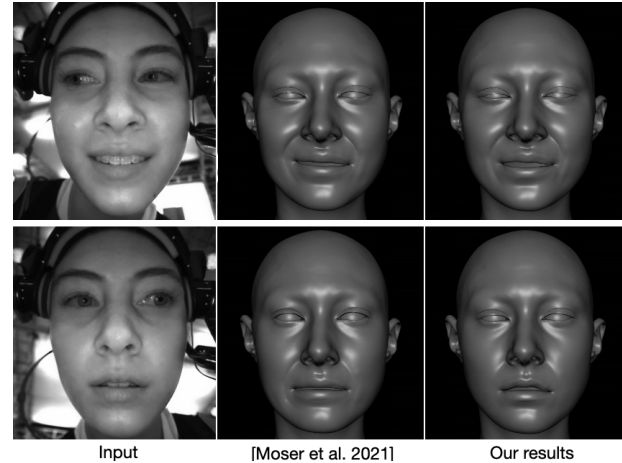


Figure 2: Comparing our results with Moser et al. [2021].

Results were captured using a HMC with 2 Ximea MQ042MG-CM cameras (60fps). Inferring images takes on average ~20ms on an Nvidia Quadro RTX5000, with the marker removal taking ~3.5sec per frame. We use Masquerade animation as groundtruth and compute a per-vertex mean error against the original work [Moser et al. 2021] and its improved version. Two examples are shown in Fig. 2. Our new results have a submillimeter error and show a general improvement of accuracy of ~10% for non-static vertices. Masquerade on the other hand takes on average ~10sec when no manual intervention/registration is needed.

The presented pipeline leverages two different approaches to achieve faster turnaround with no penalty on the quality of the animation. While it would be ideal to rely purely on our video-driven technique, the variability of the capture set still makes it unfeasible. Combining the two approaches has no impact on the recording of shots and leads to a more robust solution that evolves through a show as more shots help to refine Moser et al. [2021].

## REFERENCES

- DD. 2022. Digital Domain's Masquerade Offline Capture. Retrieved Feb 20, 2022 from <https://digitaldomain.com/technology/masquerade-offline-capture/>
- Di4D. 2022. Di4D Pro. Retrieved Feb 20, 2022 from <https://www.di4d.com/di4d-pro/>
- Disney. 2022. Anyma. Retrieved Feb 20, 2022 from <https://studios.disneyresearch.com/anyma/>
- Martin Klaudiny, Steven McDonagh, Derek Bradley, Thabo Beeler, and Kenny Mitchell. 2017. Real-Time Multi-View Facial Capture with Synthetic Training. *Computer Graphics Forum* 36 (2017).
- Lucio Moser, Chinyu Chien, Mark Williams, Jose Serra, Darren Hendler, and Doug Roble. 2021. Semi-Supervised Video-Driven Facial Animation Transfer for Production. *ACM Trans. Graph.* 40, 6 (2021).
- Lucio Moser, Darren Hendler, and Doug Roble. 2017. Masquerade: Fine-Scale Details for Head-Mounted Camera Motion Capture Data. In *ACM SIGGRAPH 2017 Talks*. New York, NY, USA, Article 18, 2 pages.
- Weta. 2022. Weta Digital - FACETS. Retrieved Feb 20, 2022 from <https://www.wetafx.co.nz/research-and-tech/technology/facets/>
- Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. 2017. Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. In *Computer Vision (ICCV), 2017 IEEE International Conference on*.